# Massyle DENDENE

PhD Candidate in Applied Mathematics
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## Research Profile

My research focuses on stochastic control of self-exciting point processes, particularly Hawkes processes, with applications to cyber risk modelling and management in finance and insurance. I study intensity control problems through dynamic programming and viscosity solutions. Alongside my research, I teach probability and financial mathematics at ENSAE Paris.

## Education

**PhD in Applied Mathematics** 2025–2028 (expected)
*École polytechnique, Institut Polytechnique de Paris – ENSAE–CREST*
*Thesis topic:* Intensity control for Hawkes processes.
*Supervisors:* Roxana Dumitrescu and Caroline Hillairet. *Co-advisor:* Nicolas Baradel.

**MSc in Probability and Finance – M2MO** 2024–2025
*Université Paris Cité, Paris*
Stochastic control and calculus, backward stochastic differential equations, partial differential equations, numerical methods and Monte Carlo simulation.

**Engineering Degree – Specialisation in Actuarial Science** 2023–2025
*ENSAE Paris, Institut Polytechnique de Paris, Palaiseau*

**Bachelor's Degree in Mathematics and Economics – MIASHS** 2022–2023
*Université de Bretagne Occidentale, Brest*

**Engineering Studies – Applied Mathematics** 2021–2023
*IMT Atlantique, Brest*

**Preparatory Classes for Grandes Écoles – PCSI-PSI*** 2019–2021
*Lycée Jean-Baptiste Say, Paris*

## Publications

## Talks and Conferences

**SIAM Conference on Financial Mathematics and Engineering** 15–18 June 2027
*Washington, DC, USA – In person*
Upcoming invited presentation in the mini-symposium on cyber risk.

**FiME PhD Seminar – Upcoming invited talk** 18 December 2026
*ISFA, Lyon, France*

**Séminaire Actuariat – Upcoming invited talk** 20 November 2026
*ISFA, Lyon, France*

## Research and Professional Experience

**Quantitative Research Intern** May–October 2025

*Exiom Partners, Paris*

- Developed a fractional jump-diffusion model for European option pricing in illiquid markets, using subordinated Lévy processes.
- Solved a fractional Dupire PIDE involving Caputo derivatives and validated the numerical scheme against Monte Carlo simulations.
- Analysed the impact of illiquidity on implied volatility surfaces and integrated the model into an economic scenario generator for life insurance.

*Actuarial dissertation:* Illiquidity risk: from European option pricing to its integration into economic scenarios for life insurance (in French).

**Admitted as a member of the Institut des actuaires following the successful defence of this dissertation.**

**Non-Life Actuarial Intern** June–September 2024

*Aon Global Risk Consulting, Luxembourg*

- Reserving for non-life reinsurance captives using Chain-Ladder, Bornhuetter–Ferguson and bootstrap methods.

**Research Intern – Supervisor: Stéphane Loisel** April–August 2023

*SAF Laboratory, ISFA, Université Claude Bernard Lyon 1*

- Developed statistical methods for longevity model selection and structural change detection in mortality data (AXA Joint Research Initiative).
- Modelled and mapped hail trajectories in Python, incorporating hailstone size data.

## Teaching Experience

**Teaching Assistant – ENSAE Paris** 2025–Present

*Institut Polytechnique de Paris, Palaiseau*

**115 hours of tutorials**, from third-year undergraduate to second-year master's level (L3–M2).

- Probability Theory.
- Financial Mathematics and Financial Instruments.
- Introduction to Stochastic Processes.

Led tutorials, supported students and graded assessments.

**Private Mathematics Tutor**

*Secondary school and classes préparatoires*

Individual support in course material and problem solving; preparation for written and oral competitive examinations, with an emphasis on rigorous reasoning and independent work.

## Skills and Languages

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|---|---|
| **Mathematics** | Probability, stochastic calculus and control, statistics, optimisation, numerical methods and time series. |
| **Programming** | Python, R, C++ (QuantLib project); Git, LaTeX. |
| **Languages** | French (native), English (C1; IELTS 7/9), German (B2), Italian (A2). |

## Interests and Activities

**Piano:** ten years of conservatory training; concerts including at the Philharmonie de Paris.

**Table tennis:** competitive play; organisation of weekly training sessions for the student club.

**Other:** climbing; regular blood donation.